MIL-I-63167(AR)
25 June 1982
SUPERSEDING
(See 6.4)

MILITARY SPECIFICATION

INSERT, NOZZLE
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for nozzle inserts used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

FEDERAL

MIL-STD-453	Inspection, Radiographic

MILITARY

MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9326204	Insert, Nozzle

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D695	Compressive Properties of Rigid Plastics, Method of Test for

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Construction. The nozzle insert shall comply with all requirements specified on drawing (dwg.) 9326204, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Materials. The nozzle insert material shall have the properties indicated in Table I. Testing shall be performed in accordance with 4.5.1, as applicable.

TABLE I. Graphite properties.

Ash content (% max.)	0.10
Apparent density (gm/cm^3)	1.60-1.70

3.3 Compressive strength. The nozzle insert shall withstand a compressive load of 5000 pounds (lbs) minimum applied axially (perpendicular to the grain flow) without cracking or breaking. Testing shall be performed in accordance with 4.5.2.

3.4 Grain flow. The nozzle insert grain flow shall be in the plane specified on the dwg. Testing shall be performed in accordance with 4.5.3.

3.5 Physical defects. The nozzle insert shall exhibit no cracks or voids of any size nor any inclusions greater than .030 inches in greatest dimension when subjected to X-Ray examination in a direction parallel to the major axis (see Fig. 1). Testing shall be performed in accordance with 4.5.4.

3.6 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.7 Workmanship. The nozzle insert shall be free of fissures, chipped or cracked edges, or any other obvious physical defects. Parts shall be clean and free of dirt, grease or any other foreign matter. Cleaning methods shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.6), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Description	Drawings	Quantity
Nozzle inserts	9326204	6
Graphite specimen for material analysis (see 3.2)	---	1

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and to the tests and inspections specified in the first article inspection table herein. All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the same source as will be used in regular production.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-63167(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1	Insert, Nozzle				9326204
					NEXT HIGHER ASSEMBLY
					M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of Defects and Tests	6			4.4.2.1
	Grain flow			3.4	
	Dimensional characteristics and marking			3.1	
	Material analysis (specimen)			3.2	
	Physical defects			3.5	
	Workmanship			3.7	
	Compressive strength (see note 1)			3.3	

NOTES: 1. These are destructive tests; perform last.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot shall consist of nozzle inserts produced from a single lot of graphite chemically analyzed in accordance with 3.2 of this specification.

4.4.2 Examinations. See MIL-A-48078 and the quality conformance inspection tables herein. Where specified sample inspection shall be performed in accordance with Table II and III, as applicable.

Table II. Sampling plan for major defects.

LQ = .01 Acceptance No. = 0 Consumers Risk = .05

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	97	93	126-127	115	162-163	137
73	72	98	94	128	116	164-165	138
74	73	99-100	95	129-130	117	166-167	139
75	74	101	96	131	118	168	140
76	75	102	97	132-133	119	169-170	141
77	76	103	98	134	120	171-172	142
78	77	104-105	99	135-136	121	173-174	143
79	78	106	100	137	122	175-176	144
80	79	107	101	138-139	123	177-178	145
81-82	80	108-109	102	140-141	124	179-180	146
83	81	110	103	142	125	181-182	147
84	82	111	104	143-144	126	183-184	148
85	83	112-113	105	145	127	185-187	149
86	84	114	106	146-147	128	188-189	150
87	85	115	107	148-149	129	190-191	151
88-89	86	116-117	108	150	130	192-193	152
90	87	118	109	151-152	131	194-195	153
91	88	119	110	153-154	132	196-197	154
92	89	120-121	111	155-156	133	198-200	155
93	90	122	112	157	134		
94-95	91	123-124	113	158-159	135		
96	92	125	114	160-161	136		

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TABLE III. Sampling plan for minor defects.

LQ = .10 Acceptance No. = 0 Consumers Risk = .10

No req'd to be delivered	Sample size for no failures
11-18	16
19-25	17
26-37	18
38-57	19
58-105	20
106-249	21
250 and over	22

Lot size required = No required to be delivered + Sample size

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Insert, Nozzle	1 OF 1	9326204
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Grain flow	4.4.3.1	---	3.4	4.5.3
102.	Outside diameter		Table II	3.1	Gage
103.	Runout of end surfaces (2 places)		Table II	3.1	Gage
104.	Largest inside diameter of diverging section (2 places)		Table II	3.1	Gage
105.	Runout of tapered inside diameter (2 places)		Table II	3.1	Gage
106.	Chamfer missing or obviously improper		Table II	3.1	Visual
107.	Surface finish of inside profile		Table II	3.1	Visual
108.	Throat inside diameter		Table II	3.1	Gage
109.	Runout of throat diameter with outside diameter		Table II	3.1	Gage
110.	Overall length		Table II	3.1	Gage
111.	Material analysis	spec.	---	3.2	4.5.1
112.	Physical defects		100%	3.5	X-Ray/4.5.4
113.	Marking missing or improper (including identifying dots)		100%	3.1	Visual
114.	Compressive strength (see note 1)		Table III	3.3	4.5.2
115.	Poor workmanship		100%	3.7	Visual

NOTES:
1. These are destructive tests.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-63167(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Packaging	1 OF 1	See 5.0
			NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements		-	5.0	Visual
102.	Objective evidence not available for packaging and packing material		-	5.1	Visual
103.	Parts not clean and free of dirt and other foreign material prior to packaging		100%	5.1	Visual
104.	Parts not individually packaged		100%	5.1	Visual
105.	Stiffening and cushioning of packages inadequate to provide firm package (when required)		100%	5.1	Visual/Manual
106.	Individual packages not properly heat sealed		100%	5.1	Visual
107.	Intermediate pack improper or not properly sealed		100%	5.1	Visual/Manual
108.	Exterior pack improper or not properly sealed		100%	5.1	Visual
109.	Marking of individual packages, intermediate packs or exterior packs missing, improper or illegible		100%	5.1	Visual
110.	Evidence of poor workmanship		100%	-	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables and herein.

4.4.3.1 Grain flow. The grain flow shall be inspected and verified by the Government quality assurance representative (QAR) on each plate of graphite in accordance with 4.5.3 before splitting to required thickness. Thereafter, the contractor shall provide the Gov't QAR with evidence of adequate controls being maintained to assure proper grain flow direction in the finished items. If the grain flow direction fails to meet the requirements of 3.4, or the controls maintained to assure proper grain flow during fabrication are inadequate, as determined by the Gov't QAR, the plate of graphite under consideration shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see MIL-A-48078 and 6.3 herein).

4.5 Test methods.

4.5.1 Materials (see 3.2).

4.5.1.1 Ash content (see 3.2). Weigh a sieved (U.S. 40) graphite sample of approximately 10 grams to the nearest 0.1 milligram into a tared 100 cc. crucible. Place the weighed sample into a muffle furnace and heat to a temperature of 800 to 900 degrees centigrade (^{o}C) (1472 to 1652 degrees farenheit (^{o}F)) for a minimum of 24 hours. Allow the crucible and sample to cool to room temperature and examine the residue for traces of carbon. If carbon is present, the ignition shall be continued until ashing is complete. When there is no further evidence of carbon, repeat the ignition for one hour periods at 800 to 900°C until a constant weight is obtained. Transfer the cooled sample to a desiccator and allow to stand for a minimum of one hour before weighing. Weigh to the nearest 0.1 milligram and calculate the ash content as indicated below. Alternate equivalent test methods may be used to perform this test when approved by the acquiring agency.

$$\text{Percent Ash} = \frac{W_1 - W_2}{W_1} \times 100$$

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Where:

W_1 = Original weight of sample
W_2 = Final weight of sample

4.5.1.2 Apparent density (see 3.2). Measure each dimension of a specimen from a sample plate of graphite in three or more locations to the nearest 1/32 inch and average each dimension. Calculate the volume of the specimen based on the average dimensions. Weigh the sample to an accuracy of .25 percent and calculate the apparent density as indicated below.

$$\text{Apparent density} = \frac{M}{V}$$

Where:

M = Weight of sample in lbs.
V = Volume of sample in cu. in.
27.68 = Conversion factor to grams per cubic centimeter

4.5.2 Compressive strength (see 3.3). The compressive strength of nozzle inserts perpendicular to the grain flow shall be tested in accordance with applicable requirements of ASTM D695. The compression specimen shall be placed on the stationery platform of the test machine and loaded at a standard head speed of .020 inches per minute until fracture occurs. Alternate equivalent test methods may be used to perform this test when approved by the procuring agency.

4.5.3 Grain flow (see 3.4). The nozzle insert material shall be visually examined for compliance with the requirements of 3.4.

4.5.4 Physical defects (see 3.5). Radiographic inspection of the nozzle inserts shall be performed in accordance with MIL-STD-453 in a direction parallel to the major axis as indicated in Fig. 1. Penetrameters shall be designed to assure a sensitivity of 2.0 percent and the film density shall be 1.50 minimum. Multiple items may be presented on a single X-Ray if positive means is provided to correlate the physical hardware with the individual nozzles presented on the X-Ray.

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5. PACKAGING.

5.1 Packaging and packing. Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with MIL-STD-1169 subject to the following details and conditions.

5.1.1 Packaging. Items shall be single item packaged in fiberboard containers, sealed in waterproof barrier bags and heat sealed in accordance with MIL-STD-1169, Level A. Barrier bags shall be fabricated from MIL-B-121 (Type II, Grade A, Class 1) material or an approved equivalent. Fillers, positioners, stiffeners and cushioning adequate to prevent physical damage to the item and to finished surfaces during transit, storage and handling shall be provided as required.

5.1.2 Intermediate packing. Single item packages shall be packed in intermediate packages in accordance with MIL-STD-1169, Level B.

5.1.3 Packing. Intermediate packages shall be packed in accordance with MIL-STD-1169, Level B.

5.1.4 Marking. Unit packages, intermediate packs and exterior containers shall be marked with the following:

a. Item nomenclature
b. Part or dwg. no.
c. Used on M753 Projectile
d. Manufacturer
e. Quantity enclosed
f. Lot no.
g. Contract or purchase order no.
h. Date of acceptance

6. NOTES

6.1 Intended use. Nozzle inserts covered by this specification are intended for use on Motor Body Bonding Assemblies for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.6).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander U.S. Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Supersession data. This specification includes the requirements of AR-PD-15, Revision A dated 29 July 1980.

6.5 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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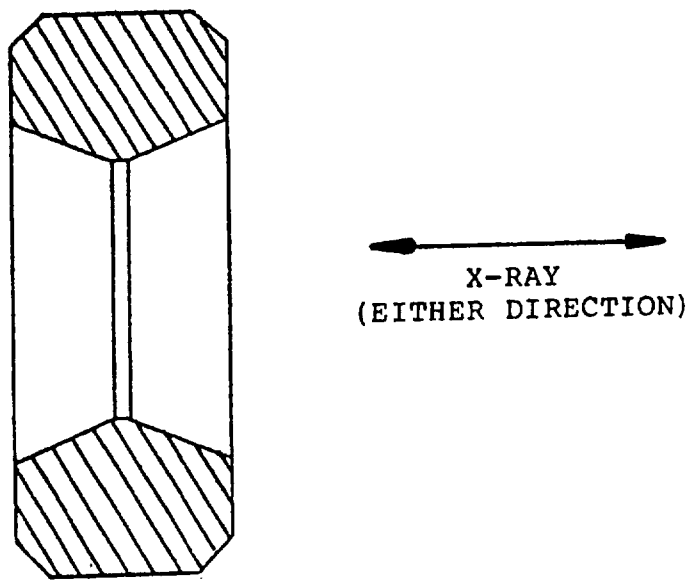


FIGURE 1. X-Ray direction (nozzle insert).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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